# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**First Semester of M. Tech. (EC) Examination**

**December-January 2013**

**EC - 703 Digital System Design**

**Date: 03.1.2014, Friday** **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

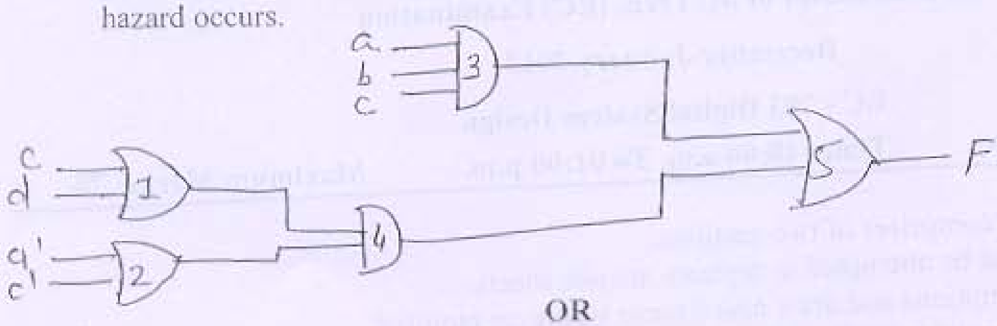
*Instructions:*

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION-I

| | | | |
|---|---|---|---|
| Q-1 | (a) | A wrong output signal produced by a defective system is called an/a _______. | 1 |
| | (b) | Which are the different types of pattern generators for BIST architecture? | 2 |
| | (c) | Construct an SM chart that has 3 input variables (D, E, F),4 output variables(P,Q,R,S), and 2 exit paths. For this block, output P is always 1, and Q is 1 if D=1. If D and F are 1 or if D and E are 0, R=1 and exit path 2 is taken. If (D=0 and E=1) or (D=1 and F=0), S=1 and exit path 1 is taken. | 3 |
| | (d) | If the initial content of the accumulator is 000001101 and the multiplicand is 1111, shows the sequence of add and shift pulses and the contents of the accumulator at each time step. | 3 |
| | (e) | A mealy sequential network with four output variables is realized using 16R4 SPLD. What is the maximum number of states for Mealy sequential network? Can any Mealy network with these numbers of inputs and outputs be realized with a 16R4. Explain. | 2 |
| Q-2 | (a) | Draw a block diagram of a 4-bit parallel multiplier. Explain operation of it and draw its control state graph. Now modify the control circuit in order to allow for a larger number of bits, so that it uses a separate counter and add shift control. Draw the state graph for the add-shift control. | 8 |
| | (b) | Compare the different programming technologies used in PLD. | 4 |
| | | **OR** | |
| Q-2 | (a) | A sequential network has one input ($x$) and one output ($z$). The output $z$=1 if the input sequence ends either in 010 or 1001, and z should be 0 otherwise. Find the Mealy state graph. | 6 |
| | (b) | What is BILBO? Describe BILBO architecture and its operation? | 6 |
| Q-3 | (a) | Draw the block diagram of Dice game, explain the SM chart for Dice game and realize it using PLA and flip-flop. | 9 |

**(b)** Find static 0 hazard in the following network. For each hazard, specify the values of the input variables and which variable is changing when hazard occurs. **3**




**OR**

**Q-3(a)** Design sequential traffic light controller for the intersection of a street "A" and "B" street with state graph. **5**

**(b)** Draw a block diagram of a 4 bit serial adder with Accumulator and Design control circuit for it. **7**

## SECTION-II

**Q-4(a)** Design 4 bit multiplier that consists of an array of AND gates and adders. **4**

**(b)** Prove that in a combinational circuit, if two faults dominate each other, then they are functionally equivalent. **3**

**(c)** Write a VHDL description for modulo 10 counter using behavioral modeling style. **4**

**Q-5 (a)** For given Boolean function below, minimize the function and generate PLA table. Implement these functions using PLA logic and draw necessary sketches. **6**

$F_1(A,B,C,D)=\sum m(2,3,5,7,8,9,10,11,13,15)$,
$F_2(A,B,C,D)=\sum m(2,3,5,6,7,10,11,14,15)$,
$F_3(A,B,C,D)=\sum m(6,7,8,9,13,14,15)$.

**(b)** Which are the main modules of keypad scanner? Explain the working of key scan using state graph. **6**

**OR**

**Q-5 (a)** Draw and explain the SM chart for UART transmitter. **6**

**(b)** Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. Also verify it with 5 variable kmap. **6**

$F(A,B,C,D,E)=\sum m(0,4,5,7,9)+\sum d(6,11)+E(m_1,m_{15})$
$F(A,B,C,D,E)=\sum m(0,4,6,13,14)+\sum d(2,9)+E(m_1,m_{12})$

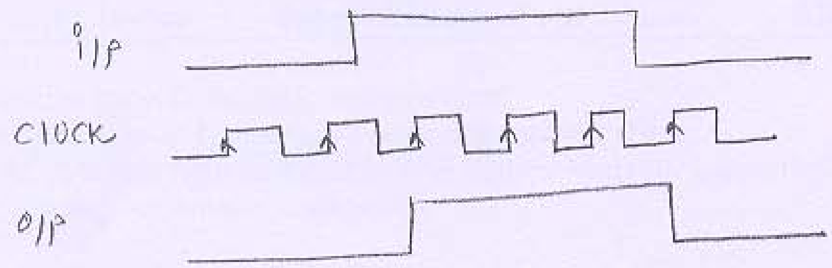
**Q-6 (a)** (i)"For combinational circuits, an undetectable fault is corresponding to a redundant wire." Justify the statement. **6**
(ii)Define: Controllling value, Inversion value

**(b)** Draw state graph and ASM chart for binary divider to divide 6 bit dividend by a 3 bit divisor to find a 3 bit quotient. The right 3 bit of the **6**

dividend register should be used to store the quotient.

**OR**

**Q-6 (a)** Design a System whose input,clock and output relationship is shown below. Write VHDL code for the same. **6**




**(b)** Consider the following circuit. Let f be the fault b s-a-0 and g be a s-a-1 **6**

a. Does f mask g under the test 0110? Does f mask g under the test 0111?